

Course: NE 533 Nuclear Fuel Performance (Beeler)-- Exam 2STUDENT'S PRINTED NAME: Ingrid Caraballo Lopez**INSTRUCTIONS FOR ADMINISTRATION:**

- To be taken **between Wednesday, February 25 and Friday, March 6, 2026.**
- Total Pages (including this cover): **Three (3)**
- Due to Engineering Online office: **end of day (11:59 pm Eastern) Friday, March 6, 2026.**
- Time limit: **75 minutes**
- **Specific Instructions:** Closed book, Closed notes, Graphing calculator allowed, No computer/internet access allowed **Equation sheet will be provided with exam. DO NOT WRITE ON THE BACK OF ANY PAGES. Additional paper needed to complete exam.**
- **Additional prohibited:** PDAs, cell phones, pagers, iPods, digital media players, text messaging devices, or other electronic devices not expressly permitted above.

THE FOLLOWING HONOR STATEMENT AND TIME RECORDING MUST BE COMPLETED:**Student - NC State University Honor Pledge:**

I, Ingrid Caraballo Lopez, have neither given nor received unauthorized aid on this exam or assignment.
(STUDENT'S PRINTED NAME)

[Signature] 8/03/26
Student Signature Date

Proctor:

Time Student Began Exam: 9:00am Time Student Completed Exam: 10:50 Total Time: 110 min.

I, Dana Wynn, administered this exam according to the instructions above.
(PROCTOR'S PRINTED NAME)

[Signature] 03/09/2026
Proctor Signature Date

RETURN INSTRUCTIONS: The completed exam (questions, answers, any work sheets/extra pages, and this form) **MUST ALL** be returned **BY THE PROCTOR** via email or fax to Engineering Online **immediately** upon completion.

- **You may scan and E-mail the exam:** Please send to examsoeol@ncsu.edu ONLY... OR
- **You may FAX the exam:** Please make a darkened photocopy of the exam and fax the darkened copy to: 919-515-9972 or 919-515-8415.

NOTE: You will be asked to mail the original completed exam. **DO NOT** mail the original of the exam until you have received an email confirmation of receipt from EOL. You will generally receive this within two business days. If you do not receive a confirmation by then, please call using our toll-free number, 1-877-254-0058 (or if local 919-515-3693), to confirm that our office has received all pages of the exam and that they are legible.

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Exam 2: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

Part A

1. (10 pts) Consider a fuel rod with a pellet radius of 0.55 cm that is experiencing a linear heat rate of 300 W/cm.
 - a. What is the maximum stress experienced by the pellet? $k_f = 0.025$ W/cm-K, $E = 200$ GPa, $\nu = 0.35$, and $\alpha = 10 \times 10^{-6}$ 1/K?
 - b. Given $\sigma_{fracture} = 175$ MPa, how far do cracks extend into the fuel?
2. (8 pts) Consider the stress state in a zircaloy-clad fuel rod pressurized to 15 MPa with an average radius of 0.55 cm and a cladding thickness of 0.05 cm.
 - a. Calculate all three components of the stress using the thin-walled cylinder approximation.
 - b. Calculate all three components of the stress at the inside wall, assuming a thick-walled cylinder.
3. (16 pts) Calculate the change in the gap thickness due to thermal expansion. Do not perform iterations. $R_f = 0.48$ cm, $t_{gap} = 0.008$ cm, $T_{CO} = 550$ K, $t_{clad} = 0.08$ cm, $k_{fuel} = 0.03$ W/cm-K, $k_{gap} = 0.003$ W/cm-K, $k_{clad} = 0.15$ W/cm-K, $LHR = 250$ W/cm, $\alpha_c = 10 \times 10^{-6}$ 1/K, $\alpha_f = 14 \times 10^{-6}$ 1/K, $T_{ref}(fuel=clad) = 300$ K.
4. (6 pts) What are the hoop and radial stresses at the inner radius of the cladding due to thermal expansion? $\Delta T = 50$ K; $\alpha_c = 15 \times 10^{-6}$; $E = 100$ GPa; $\nu = 0.34$; $t_c = 0.06$ cm; $R_i = 0.55$ cm.
5. (8 pts) Determine the densification. Fission rate = 6×10^{13} fissions/cm³-s, $\Delta \rho_0 = 0.015$, $\beta_D = 5$ MWD/kgU, $\rho(UO_2) = 10.97$ g/cc, $t = 10$ days. Assume $C_D = 1$.

Part B

6. (4 pts) Describe both elasticity and plasticity.
7. (8 pts) Define strain hardening. What causes strain hardening?
8. (3 pts) What three things must all fuel performance codes be able to do?
9. (5 pts) How do grains form in the sintering process of fuel?
10. (8 pts) Describe the concept of microstructure-based (mechanistic) fuel performance modeling and why it is beneficial.
11. (8 pts) What is meant by microstructure? Given an example of a processing technique and the impact it has on the microstructure and related properties.
12. (4 pts) Does diffusion occur more quickly inside the grains or on the grain boundaries? Why?
13. (6 pts) What is a driving force for grain growth? What can impede grain growth?
14. (6 pts) Why does fuel densify? Describe this process.

Equations for NE533

$$\begin{aligned}
 Q &= E_f \times N_f \times \sigma_f \times \phi \\
 \rho c_p \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q \\
 LHR &= \pi R^2 Q_{av} \\
 T(r) &= \frac{Q(R^2 - r^2)}{4k} + T_s \\
 T_0 - T_s &= \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k} \\
 T_s - T_{IC} &= \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}} \\
 T_{IC} - T_{OC} &= \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}} \\
 T_{OC} - T_{cool} &= \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}} \\
 h_{gap/clad} &= \frac{k_{gap/clad}}{t_{gap/clad}} \\
 k_{gap}(He) &= 16 \times 10^{-6} \times T^{0.79} \\
 k_{gap}(Xe) &= 0.7 \times 10^{-6} \times T^{0.79} \\
 k_{gap}(y) &= (1 - y) \times k_{He} + y \times k_{Xe} \\
 LHR \left(\frac{z}{Z_0} \right) &= LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \\
 T_{cool}(z) - T_{cool}^{in} &= \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \times \\
 &\quad \left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\} \\
 k_{ox} &= \frac{1}{A + BT} \text{ (W/cmK)} \\
 \frac{1}{B} \ln \left(\frac{A + BT_0}{A + BT_s} \right) &= \frac{LHR}{4\pi} \\
 A &= 3.8 + 200 \times FIMA \text{ (cmK/W)} \\
 B &= 0.0217 \text{ (cm/W)} \\
 \sigma_\theta &= \frac{pR}{\delta} \\
 \sigma_z &= \frac{pR}{2\delta} \\
 \sigma_r &= -\frac{p}{2} \\
 \sigma_r(r) &= -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1} \\
 \sigma_\theta(r) &= p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} \\
 \sigma_z(r) &= p \frac{1}{(R_o/R_i)^2 - 1} \\
 \sigma &= -E\Delta T\alpha \\
 \sigma_r(\eta) &= -\sigma^*(1 - \eta^2) \\
 \sigma_\theta(\eta) &= -\sigma^*(1 - 3\eta^2) \\
 \sigma_z(\eta) &= -2\sigma^*(1 - 2\eta^2) \\
 \sigma^* &= \frac{\alpha E(T_0 - T_s)}{4(1 - \nu)} \\
 \Delta\delta_{gap} &= \bar{R}_c \alpha_c (\bar{T}_c - T_{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref}) \\
 D(t)^2 - D_0^2 &= 2M_{GB} \gamma_{GB} t \\
 M_{GB}(T) &= M_0 e^{(-\frac{Q}{k_B T})} \\
 \frac{dD}{dt} &= k \left(\frac{1}{D} - \frac{1}{D_m} \right) \\
 \epsilon_D &= \Delta\rho_0 \left(e^{\frac{\beta \ln 0.01}{C_D \beta_D}} - 1 \right) \\
 \beta &= \frac{\dot{F}t}{N_U}
 \end{aligned}$$

Part A.

1. $r = 0.55$

$LHR = 300 \text{ W/cm}$

$k_f = 0.025 \text{ W/cm-K}$

$E = 200 \text{ GPa}$

$\alpha = 10 \times 10^{-6} \left[\frac{1}{K} \right]$

$\nu = 0.35$

$$\sigma^* = \frac{\alpha E (T_0 - T_s)}{4(1-\nu)}$$

$$\sigma_\theta(\eta) = -\sigma^* (1-3\eta^2)$$

$$\eta_1 = \sqrt{1 + \left(\frac{\sigma_f}{\frac{\sigma^*}{3}} \right)}$$

max stress

$$\eta = \frac{r}{R_f} = \frac{0.55}{\quad}$$

I don't have any more time

$$\Delta T = \frac{LHR}{4\pi k_f}$$

$$\Delta T = \frac{300}{4\pi(0.025)} = \underline{954.93}$$

$$\sigma^* = \frac{10 \times 10^{-6} (200) (\Delta T)}{4(1-0.35)}$$

$$\underline{\sigma^* = 0.734562}$$

2.
thin

$$G_{\theta} = \frac{PR}{r} = \frac{(15)(0.55)}{0.05} = 165$$

$$P = 15 \text{ MPa}$$

$$r = 0.55 \text{ cm}$$

$$r = 0.05 \text{ cm}$$

$$G_z = \frac{PR}{2r} = \frac{(15)(0.55)}{2(0.05)} = 82.5$$

$$G_r = -\frac{P}{2} = -\frac{(15)}{2} = -7.5$$

thick

$$G_r(r) = -P \frac{\left(\frac{R_o}{r}\right)^2 - 1}{\left(\frac{R_o}{R_i}\right)^2 - 1}$$

$$G_{\theta}(r) = P \frac{\left(\frac{R_o}{r}\right)^2 + 1}{\left(\frac{R_o}{R_i}\right)^2 - 1}$$

$$G_z(r) = P \frac{1}{\left(\frac{R_o}{R_i}\right)^2 - 1}$$

$$G_r(r) = -15 \frac{\left(\frac{0.575}{0.55}\right)^2 - 1}{(1.04524)^2 - 1} = -4.91814$$

$$G_{\theta}(r) = 15 \frac{\left(\frac{0.575}{0.55}\right)^2 + 1}{(1.04524)^2 + 1} = 14.2732$$

$$G_z(r) = 15 \frac{1}{(1.04524)^2 - 1} = -12.5047$$

$$R_i = R_c - \left(\frac{t_c}{2}\right)$$

$$R_o = R_c + \left(\frac{t_c}{2}\right)$$

$$R_i = 0.55 - \left(\frac{0.05}{2}\right)$$

$$R_i = 0.525$$

$$R_o = 0.55 + \left(\frac{0.05}{2}\right)$$

$$R_o = 0.575$$

$$\frac{R_o}{R_i} = 1.04524$$

3.

$$T_{ic} - T_{oc} = \frac{LHR}{2\pi R_f h_{clad}}$$

$$h_{clad} = \frac{k_{clad}}{t_{clad}} = \frac{0.19}{0.08} = 1.875 \text{ cm}$$

$$T_s - T_{ic} = \frac{LHR}{2\pi R_f h_{gap}}$$

$$h_{gap} = \frac{0.003}{0.008} = 0.375$$

$$T_o - T_s = \frac{LHR}{4\pi K_f}$$

$$K_f = 0.03$$

$$T_{oc} = 550 \text{ K}$$

$$T_{ic} = \frac{200}{2\pi(0.48)(1.875)} + T_{oc} = 585.368 \text{ K}$$

$$T_s = \frac{200}{2\pi(0.48)(0.375)} + T_{ic} = 762.207 \text{ K}$$

$$T_o = \frac{200}{4\pi(0.03)} + T_s = 1292.72 \text{ K}$$

gap calculation

$$\textcircled{1} \bar{R}_c = R_f + \delta_c + \left(\frac{t_c}{2}\right) = (0.48) + 0.06 + \left(\frac{0.06}{2}\right) = 0.57$$

$$\textcircled{2} \bar{T}_c = \frac{T_{ci} + T_{co}}{2} = \frac{585.368 + 550 \text{ K}}{2} = 567.684 \text{ K}$$

$$\textcircled{3} \Delta T_c = \bar{T}_c - T_o^{ref} = 567.684 - 300 = 267.684$$

$$\textcircled{4} \Delta R_c = \alpha_c \bar{R}_c + T_o^{ref} = (10 \times 10^{-4})(0.57) + 300 \text{ K} = 300$$

$$\bar{T}_f = \frac{(T_s + T_{co})}{2} = \frac{762.207 + 550 \text{ K}}{2} = 656.104$$

$$\Delta T_f = \bar{T}_f - T_o^{ref} = 656.104 - 300 = 356.104$$

$$\Delta R_f = \alpha_f R_f \Delta T_f = (14 \times 10^{-4})(0.48)(356.104) = 0.002393$$

$$\Delta t_g = \Delta R_c - \Delta R_f = 300 - 0.002393 = 299.997$$

$$\Delta \delta_{gap} = \bar{R}_c \alpha_c (\bar{T}_c - T_o^{ref}) - R_f \alpha_f (\bar{T}_f - T_o^{ref})$$

$$\Delta \delta_{gap} = (0.57)(10 \times 10^{-4})(567.684 - 300) - (0.48)(14 \times 10^{-4})(656.104 - 300)$$

$$\Delta \delta_{gap} = -0.000867$$

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$$3. \Delta \bar{R}_C = \alpha_C \bar{R}_C + \Delta T_C$$

$$\Delta T_C = \bar{T}_C - T_0^{\text{ref}} = 267.684$$

$$\Delta \bar{R}_C = (15 \times 10^{-6})(0.57) + 267.684 = 0.002289$$

$$\Delta R_F = \alpha_F \times R_F \times \Delta T_F$$

$$\Delta T_F = 356.104$$

$$\Delta R_F = (14 \times 10^{-6})(0.48)(356.104) = 0.002393$$

$$\Delta \bar{\rho}_g = \Delta \bar{R}_C - \Delta R_F = 0.002289 - 0.002393 =$$

$$\Delta \bar{\rho}_g = 0.000104$$

4. cladding, hoop and radial

$$\Delta T = 50K$$

$$\alpha_c = 15 \times 10^{-6} \quad E = 1006 \text{ Pa}$$

$$\nu = 0.34 \quad t_c = 0.06 \text{ cm} \quad R_i = 0.55 \text{ cm}$$

$$G_r(r) = \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{2} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$G_r(r) = \frac{1}{2} (50) \frac{(15 \times 10^{-6})(1006 \text{ Pa})}{1-0.34} \left(\frac{0.06}{0.55} - 1 \right) \left(1 - \frac{0.55}{2} \left(\frac{0.06}{0.55} - 1 \right) \right)$$

$$G_r(r) = -0.063022$$

$$G_\theta(r) = \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(1 - 2 \frac{R_i}{2} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$G_\theta(r) = \frac{1}{2} (50) \frac{(15 \times 10^{-6})(1006 \text{ Pa})}{1-0.34} \left(1 - 2 \left(\frac{0.55}{2} \left(\frac{0.06}{0.55} - 1 \right) \right) \right)$$

$$G_\theta(r) = 0.084659$$

$$5. F = 6 \times 10^{13} \text{ Atoms/cm}^2\text{-s}$$

$$\Delta p_0 = 0.015$$

$$\beta_D = 5 \frac{\text{MWd}}{\text{kgU}}$$

$$\rho(\text{UO}_2) = 10.97 \text{ g/cc}$$

$$t = 10 \text{ days}$$

$$CD = 1$$

$$1 \text{ FIMA} = 950 \frac{\text{MWd}}{\text{kgU}}$$

$$1 \text{ day} \approx 86400 \text{ s}$$

$$N_A = 6.022 \times 10^{23}$$

Number density uranium

$$n_{\text{UO}_2} = 270 \text{ g/mol}$$

$$N_U = \left(\frac{\rho(\text{UO}_2) \times N_A}{n_{\text{UO}_2}} \right) \times \left(\frac{1U}{1\text{UO}_2} \right) = \frac{10.97 \text{ g/cm}^3 \times 6.022 \times 10^{23}}{270 \text{ g/mol}} = 2.446 \text{ E}^{22}$$

Burnup of the system

$$\beta = \frac{Ft}{N_U} = \frac{(6 \times 10^{13}) (10 \text{ days} \times \frac{86400 \text{ s}}{1 \text{ day}})}{2.446 \text{ E}^{22}} = 0.000206$$

Conversion to FIMA of β_D

$$\beta_D = 5 \frac{\text{MWd}}{\text{kgU}} \times \frac{1 \text{ FIMA}}{950 \frac{\text{MWd}}{\text{kgU}}} = 0.005263$$

Strain calculation

$$\epsilon_D = \Delta p_0 \left(e^{\left(\frac{\beta \ln(0.01)}{(1)(\beta_D)} \right) - 1} \right)$$

$$\epsilon_D = 0.015 \left(e^{\left(\frac{0.000206 \ln(0.01)}{0.005263} \right) - 1} \right) = 0.004608$$

densification

Part B

6. Describing plasticity and elasticity

Plasticity is the ability of a solid material to be able to undergo a permanent deformation. It is non-reversible and this is seen in the shape of the response (typically by its applied forces). Meanwhile elasticity is the ability of a material to resist a distorting influence and return to the original shape when the influence is removed.

7. Define strain hardening

Strain hardening is caused by dislocation pile up; this can occur after plastic deformation is unloaded changing the material.

8. Fuel performance code requirements

- numerically model the temperature in the fuel
- numerically model the stress in the cladding
- Consider the gap pressure, closure and heat transfer

9. grains in sintering

can form during the heat treatment process due to the heating and cooling of the material at extreme temperatures in its powder form, without melting.

10. The concept of mechanistic based on microstructure is based on knowing the material's properties during and after a mechanical process. Since these properties obtained by the process can affect the fuel performance as imperfections to the material. It allows us to be able to predict a material's behavior.

11. microstructure

microstructure is meant by the size of the lattice structures in the material, compactness, any bubbles, pores or grains that could affect the material's performance.

- casting is an example of a process that where a liquid material is poured into a mold and allowed to solidify.
- these solidified microstructures usually are far from ideal due to its properties.

12. Diffusion/grains

It occurs faster along grain boundaries, due to that grain boundaries have more space than in the "perfect" lattice, making the atoms diffuse faster along these grain boundaries.

13. creep is the driving force of grain growth, defects in the material can reduce grain growth.

→ grain size also impacts by
fission gas release
swelling
thermal conductivity

14. Densification of fuel is when the reduction in the surface area of pores, it occurs during the sintering process as a continuation.